

Suman Kundu

Curriculum Vitae

C/O - Susanta Kundu
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Birth: 21 June, 1994

Education

- 2025–current **PostDoctoral Fellow**, *Scuola Internazionale Superiore di Studi Avanzati*, Trieste, Italy
Group head: Prof. Francesco Benini.
Expected date: 31/9/2027.
- 2022–2025 **PostDoctoral (Koshland) Fellow**, *Weizmann Institute of Science*, Rehovot, Israel
Group head: Prof. Ofer Aharony.
- 2017–2022 **Graduation (PhD)**, *Tata Institute of Fundamental Research*, Mumbai
Final Grade: 80.5
Thesis Title: Constraints on Classical Theories of Gravity.
Advisor: Prof. Shiraz Minwalla.
- Jul-Aug, '17 **Visiting Student Research Program**, *Tata Institute of Fundamental Research*, Mumbai
Project Title: Derive an action for hydrodynamics from the improved large D membrane.
Advisor: Prof. Shiraz Minwalla.
- 2015–2017 **MSc (Specialization: Gravity)**, *Indian Institute of Technology (IIT)*, Kanpur
Final Grade: 96
Thesis subject: Large D gravity and charged membrane dynamics in (A)dS.
Advisor: Prof. Sayantani Bhattacharyya.
- Dec-Jan, '15 **Visiting Project Student**, *IUCAA*, Pune
Project Title: Magnetic Field Dependence of Neutron Star Spin : Magneto-hydrodynamics.
Advisor: Prof. Kandaswamy Subramanian.
- 2012–2015 **BSc in Physics and Mathematics**, *Presidency University*, Kolkata
Final grade: 92 (Gold Medalist).

Invited Talks

- Jan, 2026 **Seminar Talk**, *IISc*, Bangalore
Subject: A String theory for 2D Yang-Mills
- Pedagogical Talk**, *IISc*, Bangalore
Subject: Aspects of large N
- Jan, 2026 **Seminar Talk**, *SNBNCBS*, Kolkata
Subject: A String theory for 2D Yang-Mills
- Dec, 2025 **Seminar Talk**, *IIT*, Gandhinagar
Subject: A String theory for 2D Yang-Mills
- Dec, 2025 **Seminar Talk**, *IISER*, Bhopal
Subject: A String theory for 2D Yang-Mills
- Pedagogical Talk**, *IISER*, Bhopal
Subject: Aspects of large N
- Pedagogical Talk**, *IISER*, Bhopal
Subject: Aspects of Confinement
- Dec, 2025 **Poster**, *Indian String Meeting 2025*, Bhubaneswar
Subject: A String theory for 2D Yang-Mills

- Nov, 2025 **Seminar Talk, SISSA, Trieste**
Subject: The S-matrix and boundary correlators in flat space
- Sep, 2025 **Colloquium, Presidency University, Kolkata, (for B.Sc. students)**
Subject: A theory of colors : Strong force
Link: <https://www.youtube.com/watch?v=vktxFi9Spg>
- Sep, 2025 **Seminar Talk, IIT, Kanpur**
Subject: A String theory for 2D Yang-Mills
- Sep, 2025 **Seminar Talk, HRI, Allahabad**
Subject: A String theory for 2D Yang-Mills
- Sep, 2025 **Seminar Talk, IISER, Pune**
Subject: A String theory for 2D Yang-Mills
- Pedagogical talk, IISER, Pune**
Subject: A String theory for 2D Yang-Mills : Mathematics behind it
- Sep, 2025 **Seminar Talk, IMSc, Chennai**
Subject: A String theory for 2D Yang-Mills
- Aug, 2025 **Seminar Talk, SINP, Kolkata**
Subject: A String theory for 2D Yang-Mills
- Aug, 2025 **Seminar Talk, IACS, Kolkata**
Subject: A String theory for 2D Yang-Mills
- Aug, 2025 **Seminar Talk, IIT Bhubaneswar, Bhubaneswar**
Subject: A String theory for 2D Yang-Mills
- Aug, 2025 **Seminar Talk, NISER, Bhubaneswar**
Subject: A String theory for 2D Yang-Mills
- Aug, 2025 **Seminar Talk, TIFR, Mumbai**
Subject: A String theory for 2D Yang-Mills
- Nov, 2024 **Seminar Talk, Tel Aviv University, Tel Aviv**
Subject: A String theory for 2D Yang-Mills
- Nov, 2024 **PCTS-Simons Collaboration Meeting, Princeton University, (15 min)**
Subject: A String theory for 2D Yang-Mills
Link: https://mediacentral.princeton.edu/playlist/dedicated/1_rsmn12h8/1_whvleaqc
- Oct, 2024 **Seminar Talk, ICTS, Bangalore, (90 min)**
Subject: A String theory for 2D Yang-Mills
Link: <https://www.youtube.com/watch?v=3VT4P9F2gE8>
- Jun, 2024 **Strings Conference, CERN, Switzerland, (Gong Show)**
Subject: String theory for 2D Yang-Mills
Link: <https://www.youtube.com/watch?v=zLH3Ma-iq-s>
- Jan, 2024 **High Energy Theory Group Meeting, Weizmann Institute of Science, Rehovot**
Subject: The S-matrix and boundary correlators in flat space
- Dec, 2023 **Seminar Talk, Technion/University of Haifa**
Subject: Towards uniqueness of Kaluza-Klein theory using CRG
- Dec, 2023 **Pre-ISM Workshop on String Theory, TIFR, Mumbai**
Subject: ‘Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability
- Nov, 2023 **Seminar Talk, TIFR, Mumbai**
Subject: Regge growth of isolated massive spin-2 particles and the Swampland
- Aug, 2023 **Seminar Talk, Technion/ University of Haifa**
Subject: ‘Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability
- Jul, 2023 **PITP, IAS, Princeton, (Poster)**
Subject: ‘Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability

- Jul, 2023 **General Relativity Seminar, Centre for Mathematical Science and Applications, Harvard University**
 Subject: ‘Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability
 Link: https://cmsa.fas.harvard.edu/event/gr_7523/
- Jun, 2023 **High Energy Theory Group Meeting, Weizmann Institute of Science, Rehovot**
 Subject: ‘Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability
- Jan, 2023 **Joint Seminars in Theoretical High Energy Physics, Neve Shalom**
 Subject: CRG from bound on chaos
- Jul, 2022 **Strings Conference, University of Vienna, (Poster)**
 Subject: Classification of Monodromies in Lorenzian CFT correlators in 2D
- Dec, 2021 **Indian String Meeting, IIT, Roorkee, (Introductory talk)**
 Subject: CRG from bound on chaos
 Link: <https://www.youtube.com/watch?v=vIOgRZamxxA>
- Dec, 2021 **Seminar Talk, Greater Paris Bootstrap Collaboration (IPhT Saclay, Ecole), Paris**
 Subject: CRG from bound on chaos
- Nov, 2021 **Seminar Talk, Caltech**
 Subject: CRG from bound on chaos
- Aug, 2021 **Seminar Talk, IAS Princeton**
 Subject: CRG from bound on chaos
- Jul, 2021 **TASI 2021, UC Colorado Boulder, (Gong show)**
 Subject: CRG from bound on chaos
- Jun, 2021 **Strings 2021 conference, ICTP-SAIFR, (Gong Show)**
 Subject: CRG from bound on chaos
 Link: <https://youtu.be/Cv92qxKt2LA?t=1787>
- Sep, 2020 **Colloquium, Presidency University, Kolkata, (for B.Sc. students)**
 Subject: CRG from bound on chaos
 Link: <https://www.youtube.com/watch?v=lp1ti5YRZHo>
- 2018 **Colloquium, NISER, Bhubaneswar**
 Subject: Large D gravity and charged membrane dynamics with nonzero cosmological constant.

Teaching Experience

- 2026 **Set of three lectures: large N , Confinement, SUSY Localization, IISER Bhopal, IISc Bangalore**
 three classic ideas as prerequisite for understanding non-perturbative string-dual picture of QCD.
- 2023 **Mentored in Research project, TIFR**
 Students: Asikur Rahman, Vipul Kumar (2311.17015), Abhishek Navhal (2505.01507)
- 2022 **Tutorial in Kavli Asian Winter School, ICTS**
 Instructor: Prof. Shiraz Minwalla
 Link: <https://www.youtube.com/watch?v=olOb-pH0NO8>
- 2022 **TA for Super Symmetry course, TIFR, Mumbai**
 Instructor: Prof. Shiraz Minwalla
- 2021 **TA for Electrodynamics II course, TIFR, Mumbai**
 Instructor: Prof. Sushil Muzumdar
- 2020 **TA for Advanced Quantum Mechanics course, TIFR, Mumbai**
 Instructor: Prof. Rishi Sharma
- 2019 **TA for Group Theory and Topology course, TIFR, Mumbai**
 Instructor: Prof. Goutam Mandal

Awards and Achievements

- 2023 **The Koshland Prize**, *Weizmann Institute of Science*
Awarded for exceptionally outstanding Postdoctoral Fellows at the Weizmann Institute and for up to three years, an augmented monthly fellowship that is 25% higher than the regular WIS fellowship.
- 2021 **ICTP Fellowship**, *ICTP, Trieste*
Awarded first prize for presenting a gong show in ‘Strings 2021’ from developing countries.
- 2016 **Dr. Shyama Prasad Mukherjee (SPM) Fellowship**, *CSIR, India*
awarded through interview after securing 4th rank in CSIR-UGC JRF (NET) in 2016. (worth 1,00,000 INR per year)
- 2015 **Makhan Lal Sarkar Memorial Prize**, *Presidency University, Kolkata*
awarded for securing First position in B.Sc. as total grade of the Presidency University.
- 2013 **INSPIRE Scholarship**, *Ministry of Human Resource Developments, Government of India*
awarded on the basis of performance in (10+2) equivalent exam (worth 60,000 INR per year)
- 2009 **Merit-cum-means Scholarship**, *West Bengal Govt.*
- 2015 **First rank in B.Sc.**, *Presidency University*
- 2015 **Second rank in JEST**, *Int-PhD selection exam in all India*
- 2015 **Third rank in JAM**, *Int-PhD selection exam in all India*
- 2015 **Fourth rank in NET**, *National PhD selection exam in all India*
- 2015 **Subject (Physics) GRE Scaled Score- 940**, percentile: 90

Conferences

- Dec, 2025 **Indian String Meeting**, *Bhubaneswar, India*
- Nov, 2025 **Simons QCD meeting**, *New York, USA*
- Nov, 2024 **Simons QCD meeting**, *New York, USA*
- Jul, 2024 **Simons QCD meeting**, *Les Diablerets, Switzerland*
- Jun, 2024 **Strings 2024**, *CERN, Switzerland*
- Dec, 2023 **Indian String Meeting**, *IIT Bombay*
- July, 2022 **Strings 2022**, *University of Vienna*
- Dec, 2021 **Indian String Meeting (Online)**, *IIT, Roorkee*
- Jun, 2021 **Strings 2021 (Online)**, *ICTP-SAIFR, Sao Paulo*
- Jun, 2020 **Strings 2020 (Online)**, *University of Cape Town*
- Dec, 2019 **National String Meeting**, *IISER, Bhopal*
- Dec, 2018 **Indian String Meeting**, *IISER, Trivandam*

Schools/Workshops

- Nov, 2025 **Simons Confinement workshop**, *Simons Center at Stony Brook, USA*
- May, 2025 **Bootstrap, Holography & Swampland: Imprint of UV principles on IR physics**, *Aspen Center for Physics, Colorado*
- Nov, 2024 **PCTS-Simons Confinement workshop**, *Princeton University, New Jersey*
- Dec, 2023 **String Theory Workshop**, *TIFR, Mumbai*
- Jul, 2023 **PITP school**, *IAS, Princeton*
- Dec, 2022 **Jerusalem Winter School**, *The Hebrew University*
- Aug, 2021 **Superstring Theory and Related Topics (Online)**, *ICTP (Italy)*
- Jul, 2021 **Recent Developments in S-matrix theory (Online)**, *ICTS (Bengaluru)*
- Jul, 2021 **ST⁴ (Online)**
- Jun, 2021 **TASI school**, *UC Colorado Boulder*
- Jun, 2021 **Pre-Strings school (Online)**, *IIP, Natal, Brazil*

Feb, 2021 **CERN Winter School on Supergravity, Strings and Gauge Theory (Online)**
 Feb, 2021 **Bootstrap Lectures (workshop)**, IISc, Bengaluru
 Jan, 2021 **Kavli Asian Winter School on Strings-Particles and Cosmology**, Tohoku University
 Nov, 2020 **Strings and Fields 2020**, Kyoto University
 Sep, 2019 **KIAS Autumn Symposium on String Theory**, KIAS, Seoul

Skills

Language English, Bengali (Native language), Hindi.
 Numerics Mathematica, C, C++, Python, FORTRAN, L^AT_EX.
 2024–2025 Worked as a Mentor of final year PhD students as a part of the initiative ‘Quantum Guides’.
 Link: <https://sites.google.com/view/mentoring-program-string-th/home?authuser=0>
 2023–2025 Arranging Indian Cultural program at Weizmann Institute as part of diversity initiative
 2017–2022 Active member of Journal Club in TIFR.
 2013–2015 Active member of Presidency Physics League, Kolkata.
 2017–current Alumni of IIT Kanpur.
 2020–2021 Head of TIFR canteen monitoring committee.
 2020–2022 Department representative in TIFR COVID task force group.

Referees

- Prof. Shiraz Minwalla, DTP, TIFR, Mumbai,
 email: shiraz.minwalla@gmail.com,
 Relationship: Ph.D. Thesis Advisor and Course instructor for Advanced courses (String theory, Classical gravity, and QFT II)
- Prof. Ofer Aharony, DPPA, Weizmann Institute, Israel
 email: ofer.aharony@weizmann.ac.il
 Relationship: Post doctoral Advisor and Collaborator
- Prof. Ashoke Sen, ICTS-TIFR, Bengaluru
 email: ashokesen1999@gmail.com
 Relationship: Discussed my research in multiple occasions.
- Prof. Eran Palti, String Theory Dept, Ben Gurion University of the Negev, Israel
 email: palti@bgu.ac.il
 Relationship: Collaborator
- Prof. Abhijit Gadde, DTP, TIFR, Mumbai
 email: abhijitgadde@gmail.com
 Relationship: Collaborator

Disclaimer

- Number of published papers: 10; Number of papers with ongoing review process: 2.
- In ‘high energy Physics - theory’ the publications authors are ordered according to surname.
- ‘ * ’ marked papers are submitted to arxiv, journal for publication. Review process is ongoing.
- ‘ ** ’ marked papers are accepted in journal for publication but not yet surfaced.

Publications

*Offer Aharony, Suman Kundu, and Tal Sheaffer. A String Theory for Two-Dimensional Yang-Mills Theory II, ArXiv:2510.22303. 10 2025.

Suman Kundu, Shiraz Minwalla, and Abhishek Navhal. Monodromies of CFT correlators on the Lorentzian cylinder. *SciPost Phys.*, 19(6):162, 2025, Impact Factor: 4.85.

Offer Aharony, Suman Kundu, and Tal Sheaffer. A string theory for two dimensional Yang-Mills theory. Part I. *JHEP*, 07:063, 2024, Impact Factor: 5.5.

Subham Dutta Chowdhury, Vipul Kumar, Suman Kundu, and Asikur Rahaman. Regge constraints on local four-point scattering amplitudes of massive particles with spin. *JHEP*, 05:123, 2024, Impact Factor: 5.5.

**Diksha Jain, Suman Kundu, Shiraz Minwalla, Onkar Parrikar, Siddharth G. Prabhu, and Pushkal Shrivastava. The S-matrix and boundary correlators in flat space, ArXiv: 2311.03443. Update: Accepted for publication in JHEP.

Suman Kundu, Eran Palti, and Joan Quirant. Regge growth of isolated massive spin-2 particles and the Swampland. *JHEP*, 05:139, 2024, Impact Factor: 5.5.

Seok Kim, Suman Kundu, Eunwoo Lee, Jaeha Lee, Shiraz Minwalla, and Chintan Patel. Grey Galaxies’ as an endpoint of the Kerr-AdS superradiant instability. *JHEP*, 11:024, 2023, Impact Factor: 5.5.

Sachin Jain, Nilay Kundu, Suman Kundu, Abhishek Mehta, and Sunil Kumar Sake. A CFT interpretation of cosmological correlation functions in α -vacua in de-Sitter space. *JHEP*, 05:111, 2023, Impact Factor: 5.5.

Deeksha Chandorkar, Subham Dutta Chowdhury, Suman Kundu, and Shiraz Minwalla. Bounds on Regge growth of flat space scattering from bounds on chaos. *JHEP*, 05:143, 2021, Impact Factor: 5.5.

Soumangsu Chakraborty, Subham Dutta Chowdhury, Tushar Gopalka, Suman Kundu, Shiraz Minwalla, and Amiya Mishra. Classification of all 3 particle

S-matrices quadratic in photons or gravitons. *JHEP*, 04:110, 2020, Impact Factor: 5.5.

Suman Kundu and Poulami Nandi. Large D gravity and charged membrane dynamics with nonzero cosmological constant. *JHEP*, 12:034, 2018, Impact Factor: 5.5.

Yogesh Dandekar, Suman Kundu, Subhajit Mazumdar, Shiraz Minwalla, Amiya Mishra, and Arunabha Saha. An Action for and Hydrodynamics from the improved Large D membrane. *JHEP*, 09:137, 2018, Impact Factor: 5.5.